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SUPERSEDING
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17 March 1961

MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, DELAY, M44 ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, Delay, M44, Assembly, loaded with Cartridge, Initiator, Delay M114 for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MS28741	- Hose Assembly, Detachable End-Fitting, Medium Pressure

DRAWINGS

Munitions Command

F10520228	- Initiator, Cartridge Actuated, Delay, M44, Assembly

FSC 1377

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2.1 DRAWINGS (Cont'd)

Munitions Command

C8796826	- Container, Ammunition, Fiber, M375 for Initiators, Delay, M4, M6A1, M12, M26, M33, M44, M45 or M49 (See 5.1)
C8796827	- Box, Packing, Ammunition for Initiators, Delay, M4, M6A1, M12, M26, M33, M45 or M49 in Fiber container M375 (See 5.1)
IEL-10520228	- Inspection Equipment List for Initiator, Cartridge Actuated, Delay, M44, Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Breakaway torque.- The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.3 Residual magnetism.- The initiator shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional

3.4.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be recorded at the end of 3 feet of aircraft hose, size No. 4, in accordance with MS 28741.

TABLE I

Temperature	Pressure (PSI)	Ignition Delay (See 6.2)
-65° to -70°F	1300 to 2200	4.20 seconds, max.
65° to 75°F	1300 to 2200	3.00 plus or minus 0.3 sec.
160° to 165°F	1300 to 2200	2.30 seconds, min.

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3.4.1.1 Initiation.- The initiator shall be fired by a source delivering an output equal to that of an M3A2 initiator at the end of 15 feet of aircraft hose, size No. 4 (in accordance with MS 28741) at 70° ± 5° F.

3.4.2 The initiator shall not fail to fire.

3.4.3 The initiator shall function without mechanical failure (See 6.1).

3.5 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submittal of product.- The contractor shall submit the product in accordance with MIL-STD-105.

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4.2.1.2 Size of lot.- A lot shall consist of an indentifiable group of assembled initiators containing metal parts assemblies produced by a homogeneous process and loaded with accepted cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plan and criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Initiator, Delay, M114, Drawing D8596110 covering a detail of Drawing F10520228.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. "O" ring missing	Visual
4. Sealing disc missing, damaged or improperly secured	Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. "O" Ring split, cut or cracked	Visual
102. Foreign matter	Visual
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Varnish missing or inadequate	Visual

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4.2.2.2 Assembly, Drawing F10520228, less Drawings B8594745, D8595254, D8593313, D8593314 and D8596110.

Defect	Method of Inspection
CRITICAL	
1. Protrusion of firing pin	Gage
2. "O" ring missing	Visual
MAJOR	
None defined	
MINOR	
None defined	

4.2.2.3 Assembly, Drawing F10520228, less Drawings D8595254, D8593313, B8593314, D8596110, B8593298 and B8359536.

CRITICAL	
1. Shear pin not engaged in firing pin	Gage
2. Shear pin set screw missing	Visual
MAJOR	
101. "O" ring missing or not lubricated	Visual
102. Thread damaged	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Assembly Drawing F10520228, less Drawings D8595254, D8593314, B8359536 and B8593298.

CRITICAL	
1. Cartridge loose	Visual and Manual
2. Cartridge retainer loose	Visual and Manual
3. Metering slot of cartridge retainer	Visual
4. Elongated slot of cartridge retainer missing	Visual

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4.2.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
None defined	

4.2.2.5 Assembly, Drawing F10520228, Less Drawings A8359536 and B8593298

CRITICAL	
1. Cartridge retainer position, improper	Gage
2. Foreign matter in gas port	Visual
MAJOR	
101. Length from cap boss face to gas port face, max.	Gage
102. Set screw missing, loose or above flush	Visual and Manual
MINOR	
None defined	

4.2.2.6 Assembly, Drawing F10520228.

CRITICAL	
1. Base of firing pin housing not in complete contact with face of cartridge	Manual
2. Any part missing, except as other classified	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Height of cap, max.	Gage
102. Set screw not covered with barrier material	Visual
MINOR	
201. Total length, max.	Measure
202. Location of labels, incorrect	Visual
203. Shipping cap or plug missing	Visual
204. Protective finish inadequate or defective	Visual

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4.2.2.7 Fiber Container, Drawing C8796826.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damage through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.8 Packing box, Drawing C8796827

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual and Manual
203. Strapping missing, loose, broken or improperly assembled	Visual and Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

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4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.- Each initiator shall be subjected to the torque application. Failure of any initiator assembly to comply with the requirements of 3.2 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism.- Each initiator shall be subjected to the test for excessive residual magnetism. Failure of any initiator to comply with the requirements of 3.3 shall be cause for rejection of the initiator.

4.2.3.3 Functional.

4.2.3.3.1 Ballistics.- Thirty units shall be tested to determine compliance with the requirements of 3.4.1.3.4.2 and 3.4.3. The units shall be temperature conditioned as provided in Table III and 4.3.3.1.1. Failure of any unit to comply with the requirements of 3.4.1, 3.4.2 and 3.4.3 shall be cause for rejection of the lot.

TABLE III

Temperature	Sample Size
-65° to -70°F	10
65° to 75°F	10
160° to 165°F	10

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10520228.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.- A torque wrench set at the torque requirement specified in 3.2 shall be used in conjunction with a fixture to determine the adequacy of the joints. The torque shall be applied to the cap.

4.3.2 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90° rotation of the initiator.

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4.3.3 Functional.

4.3.3.1 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample unit shall be reconditioned in accordance with 4.3.3.1.1.

4.3.3.1.1 Temperature conditioning.- A standard M44 initiator, loaded with a dummy M114 Cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8796826 and C8796827, except that marking requirements shall reference M44 initiator.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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